

PROCLITICIZED PHONOLOGICAL PHRASES IN ENGLISH: EVIDENCE FROM RHYTHM*

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Abstract. Post-lexical stress shift in English, as exemplified by *FIFteen MEN* (cf. *fifTEEN* or *FIFTEEN*) is sensitive to rhythm and constituency. Rhythm can be accounted for by constraints like *NoClash*, forbidding adjacent accents within the phonological phrase. Constituency has been accounted for by a cyclic application of right-alignment and left-alignment of accents in the phonological phrase. However, such an account fails to take care of multiply premodified NPs like *Fifteen Japanese CDs*. Ranking right-alignment above left-alignment gives *fifTEEN japaNESE ceeDEES* and the reverse ranking gives the equally unacceptable *FIFteen JAPanese CEEdees*. Abandoning a cyclic treatment, I propose a procliticized structure in which procliticized phonological phrases correspond to embedding premodifiers like *fifteen* in the above example. Such premodifiers align with the left edge with a phonological phrase but fail to align their right edge with a phonological phrase. As a result, the correct (*FIFteen (JAPanese ceeDEES)*) is produced, or, if *NoClash* is given a stricter interpretation so as to ban accents with only one unaccented syllable between them, the equally correct (*FIFteen (JAPanese ceeDEES)*). A systematic but hitherto unnoticed stress contrast is discussed, that between e.g. *TOO high BUILDings* and *TWO HIGH BUILDings*, and it is shown that the new account deals with it effortlessly.

1. Introduction

Stress shift data in English of the type illustrated by *FifTEEN* vs *FIFteen MEN* and *OLD* vs *VERY old MEN* (Chomsky & Halle 1986: 117) have attracted considerable attention. One issue has been the phonological representation of the phenomenon. Early ‘non-linear’ accounts tended to view it as a change in the level of stress on syllables, following Chomsky & Halle’s ‘linear’ phonological tradition, but couching descriptions in terms of metrical grids or metrical trees (Liberman & Prince 1977; Selkirk 1984). Later accounts, however, have tended to take shifted and non-shifted pronunciations to represent different patterns of pitch accent distribution (Gussenhoven 1986, 1991; Ladd & Monaghan 1987; Shattuck-Hufnagel 1989; van der Hulst 1999), adopting an earlier similar position (Bolinger 1965, 1986; Vanderslice & Ladefoged 1972). The renewed claim that English clash resolution amounts to an arrangement of pitch accents, usually a deletion of the leftmost of the two clashing pitch accents, received support from acoustic measurements (Horne 1990;

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Grabe & Warren 1995; Vogel, Bunnell, & Hoskins 1995), which have tended to show that the syllable to which stress is 'shifted' often shows little or no phonetic variation between shifted and non-shifted pronunciations, while the syllable from which the stress is 'shifted' does. This suggests that the most general difference between the 'non-shifted' and 'shifted' pronunciations is the absence of a pitch accent on the syllable with main word stress in the 'shifted' pronunciation. Another source of evidence was provided by data from the Boston University Radio News Corpus (Shattuck-Hufnagel 1995; Shattuck-Hufnagel, Ostendorf, & Ross 1979; Ostendorf, Price, & Shattuck-Hufnagel 1995). In this paper, I adopt the view that the data indeed concern patterns of pitch accent distribution, indicating accented syllables by capitalization, as is customary.

The second issue concerns the role of the morpho-syntactic structure. It has long been evident that the data are not determined by rhythm alone. For instance, in addition to an Alternation Principle, Selkirk (1984) proposed that in longer NPs, the morpho-syntactic bracketing determined the surface patterns. To account for this fact, she presented a cyclic treatment of complex NPs like *ALEwife Brooke PARKway SUBway station*. The aim of this paper is to show that a cyclic solution indeed gets close to producing the correct surface forms, but that it is systematically incapable of accounting for multiply premodified NPs like *FIFteen JAPANese conSTRUCTIONS*. To account for all the data considered here, I argue that a phonological phrase structure is required with procliticized phrases corresponding to embedding premodifiers like *fifteen* in the above example.¹

Before the problem is explained and a solution is offered, it is convenient to distinguish two aspects of the morpho-syntactic structure that play rather different roles. First, rhythmic readjustments of accents interact with morphological constraints on accent locations (Prince 1983). Consider the difference between (1a) and (1b). Both NPs contain four words, but differ in the location of the first accent. It might at first sight be thought that this is due to a difference in bracketing, since (1a) has the structure $[[A B][C D]]$, while (1b) is $[[[A B] C] D]$. (NB. The assumed pronunciation for the phrase *Big Band* in isolation is *BIG BAND*, like *HUDson BRIDGE*, not *BIG band*, like *HUDson Street*.) However, this cannot be the whole answer, since (1c), my example, is also $[[[A B] C] D]$, yet patterns like (1a).

- (1) a. TOM Paine Big BAND (i.e., a Big Band led by Tom Paine)
- b. Tom PAINE Street BLUES (i.e., blues induced by Tom Paine Street)
- c. TOM cat free ROOF (i.e., a roof free of tomcats)

¹ Schreuder & Gilbers (2004) independently make a proposal for procliticized ϕ 's in multiply premodified NPs in Dutch, on the basis of phonetic data.

Second, premodified NPs of the type (2a) are distinct from structures like (2b) (Gussenhoven 1991).

- (2) a. with GROWing CHINEse supPORT
 ‘with Chinese support which is growing’
 b. with ETHnic ChiNESE supPORT
 ‘with support from the ethnic Chinese population’

Respectively, these aspects concern lexical and post-lexical generalizations about pitch accent distributions in English. Our problem concerns the post-lexical part, i.e. the account of data like (2). Nevertheless, for a proper understanding of the problem, a treatment of the lexical data is called for.

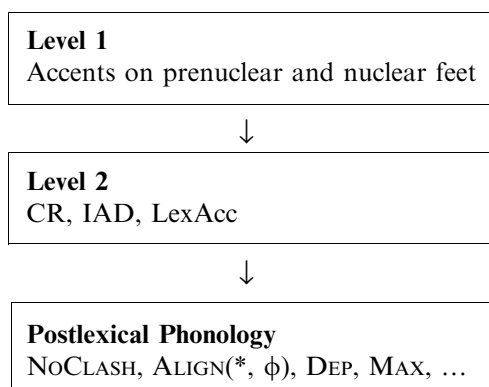
The account is presented in Optimality Theory (Prince & Smolensky 1993; McCarthy & Prince 1993, 1995; Kager 1999; McCarthy 2002), in a version that allows the direct expression of generalizations that are exclusively valid at either the lexical or the post-lexical levels, and within the lexical component at separate levels of the lexicon (Kiparsky 2000, 2002). In (3), each box represents a module with an OT grammar. In the first two modules, morphological operations and the accompanying phonological adjustments take place, Level 1 and Level 2.² At level 1, underived words like *CELLar*, *ELephant* and *karaOke* receive their lexical stress patterns. Certain suffixation processes, too, take place here, which reveal themselves through affecting the position of the lexical stress. For instance, the Level-1 suffixes *-ity* and *-ic* cause shifts in the main stress from *SEnile* to *seNILity*, and from *STRATegy* to *straTEgic* (cf. Burzio 1994; Zonneveld, Trommelen, Jessen, Rice, Bruce, & Árnason 1999). Level-1 output forms include (a) feet, (b) markers for main stress, and (c) accents on all feet except those after the main stress (Gussenhoven 1994).

The output of Level 1 serves as the input to Level 2. Only syllables that are accented in Level 1 output forms can appear as accented in surface representations, and all subsequent generalizations therefore involve accent deletions. The output of the second grammar, Level 2, has undergone accent deletions as a result of two constraints, the COMPOUND RULE (CR) and INITIAL ACCENT DELETION (IAD). A further relevant constraints here is LEXACC, which requires every Level 2 output to have at least one accent. Their workings are the subject of the next section.

² Level 3, not included in (3), is reserved for affixations which leave the prosodic structure of the base intact, among which are inflections like plural [z], past [d] and numeral and comparative suffixes.

The third module is postlexical. Here NoCLASH is active, along with constraints aligning the edges of the phonological phrase (ϕ) with accent and constraints that demand a faithful reproduction of the accentuation of the input, the Level 2 output forms. The postlexical grammar is the main topic of the article, and is dealt with in section 2.

(3)



1.1. *Deaccentuation in the lexicon*

This section aims to account for the output of the lexicon (the *lexical representations*, in the sense of Kiparsky 1982), or equivalently, the input to the postlexical module. In the view defended here, the expressions in (1) differ because they present themselves to the postlexical grammar of English in different guises: they leave the lexicon with different accent distributions.

The first opportunity for the deletion of the accents that were supplied at Level 1 is at level 2. Here, compounds are formed, like those in (4a), as well as suffixed forms whose main stress is as in the base, as in *STRATegy* – *STRATegist*; *GENTleman* – *GENTlemanliness*, which have one or more of the suffixes listed in (4b). To these, we should add similarly ‘stress-neutral’ suffixes like those in (4c), occurring in *COLourfast*, *STRIKE-prone*, *MONey-wise*, *ROADworthy*, etc. These are semantically and prosodically weightier, but are unaccented, like the suffixes in (4b).

(4) Morphological operations at level 2

- a. *language conference*, *book exhibition*, *highchair*
- b. *-ish*, *-ist*, *-ly*, *-less*, *-ness*
- c. *-fast*, *-free*, *-proof*, *-prone*, *-style*, *-tight*, *-type*, *-wise*, *-worthy*

The COMPOUND RULE (CR) deletes accents in righthand constituents of compound words, here given as constraint (5). Any accent in the

righthand constituent constitutes a violation of CR. As a result of its high ranking, compounds will survive with accents in the left-hand constituent only, and the compounds in (4) are thus pronounced *LANGuage conference*, *BOOK exhibition*, *HIGHchair*. The non-accentuation of the suffixes in (4c) could either be explained by their status as suffixes, so that they are treated like (4b), or by assuming they are treated as righthand constituents of compounds, i.e. as falling under (4a).

The second constraint operative at Level 2 is INITIAL ACCENT DELETION (6) (Gussenhoven 1991). IAD causes all accents except the rightmost to be deleted in any formation listed in (4). In OT, IAD is an alignment constraint, requiring accent to align with the right edge of a Level-2 formation. Thus, every non-final accent violates IAD by as many accentable positions (= stressed syllables) as it is removed from the right edge. A separate constraint is required to prevent a completely unaccented lexical representation (i.e. input form to the postlexical module), LEXACC (7). The high ranking of LEXACC, CR and IAD is specific to Level-2 morphology: neither has anything to say about other structures. These data support Kiparsky's level-ordered OT grammar, where sequentially ordered subgrammars, each containing their own constraint ranking, produce intermediate representations of the kind envisaged here.

- (5) COMPOUND RULE (CR) (Level 2): The righthand constituent of compound words is unaccented
- (6) INITIAL ACCENT DELETION (IAD) (Level 2): Align accent with the right edge of a Level-2 formation
- (7) LEXACC (Level 2): A lexical representation is accented

The treatment of *Tom Paine Street* and *tomcat-free* is shown in tableau (8). Winning candidate *Tom PAINE Street* maximally satisfies IAD without violating CR. In the case of *tomcat-free*, we are dealing with a phrasal formation, in which two lexical representations are joined in an Adjective Phrase. Both constituents are therefore treated separately at Level 2. Candidate (a) is better than candidate (b), due to CR. A fully unaccented *tomcat* is ruled out, since at the end of the lexicon, LEXACC requires an accented expression. This constraint rules out candidate (c), which violates LEXACC in its first constituent, and candidate (d), which violates it in both *tomcat* and *free*. Without a tableau, it should be clear that the phrases *Tom Paine* and *Big Band* in *Tom Paine Big Band* cannot violate CR or IAD, and that since *Tom Paine Big Band* is a phrase, this expression enters the postlexical phonology as a fully accented *TOM PAINE BIG BAND*.

(8)

	LEXACC	CR	IAD
[[[TOM PAINE] STREET] _{Level2}			
☞ a. tom PAINE street			*
b. TOM paine street			**!
c. TOM PAINE street			**!* **!
d. TOM paine STREET		*!	**
e. tom paine STREET		*!	
[TOM CAT] _{Level2} [FREE] _{Level2}			
☞ a. TOM cat FREE			*!/-
b. tom CAT FREE		*!/-	
c. tom cat FREE	*!/-		
d. tom cat free	*!/*		

The assumption that after Level 1 accents cannot be introduced implies that DEP-IO(Acc), which forbids gratuitous addition of accents, is undominated (9). As a result, *-wise*, etc., will not be accented (cf. (4)). The MAX constraint (10), by contrast, is ranked below the constraints discussed so far. We will see it in action in the next section.

(9) DEP-IO(Accent): Do not insert accents

(10) MAX-IO(Accent): Do not delete accents

As a result of IAD, prenuclear pitch accents are contrastive in a phrase like *SECond LANGuage conference* ‘second of a series of language conferences’, where both *second* and *language conference* are accented. The expression contrasts with the compound *second LANGuage conference* ‘conference on the topic of second language acquisition’. Similarly, a phrasal adjective like *UN-KIND* is formed in the same way as *ANGlo-aMERican*, *PROto-GerMANic*, and thus pronounced with two pitch accents. By contrast, the word *unKINDness*, which is a Level 2 formation, has a pitch accent on *kind* only when spoken in isolation (Gussenhoven 1991), just as *Proto-GerMANic teacher* ‘teacher of Proto-Germanic’ only has one on *-MAN-*.

After this section, which essentially recasts the analysis of Gussenhoven (1991) in OT terms, we move on to the post-lexical generalization, NOCLASH.

2. Postlexical rhythm: ϕ -structure

After showing the competition between NOCLASH and the edge-alignments of accents in section 2.1, I first demonstrate in section 2.2

that the data in (2) can indeed be accounted for in a cyclic fashion, as envisaged by earlier analyses, and show how the same analysis can be expressed in OT with the help of Output-Output faithfulness. Then, in section 2.3, I present data that fail to yield to this cyclic analysis: multiply premodified NPs. These data show a bias towards initial pitch accents in internal constituents, seemingly requiring high-ranking NoCLASH to be demoted below the left-alignment of accents. Section 2.4 shows how of a number of possible responses, only a proclicized ϕ -structure appears to produce the correct results, as these [A B [C D]] structures can be provided with accent on ϕ -initial C without requiring preceding B to be accented, since B is neither ϕ -final nor ϕ -initial. Finally, in section 2.5, I show how the account naturally extends to a so far undiscussed contrast in English, that between NPs with adverbs in adjective phrases, [[Adv Adj] Noun] and NPs with with multiple premodifications, like [Adj Adj [Noun]]. A conclusion is offered in section 3.

2.1. *The basic analysis*

It has long been understood that the crux to post-lexical rhythmic adjustments lies in the interaction between a tendency towards alternating rhythm, or, in an alternative formulation, an avoidance of adjacent prominences, and the prosodic structure, specifically the phonological phrase (ϕ) (Selkirk 1978; Nespor & Vogel 1986; Gussenhoven 1991). Prince's End Rules (Prince 1983) and Selkirk's edge-based account of syntax-phonology mapping (Selkirk 1986) gave rise to the edge-alignment constraints in OT that will play an important role in the present account (McCarthy & Prince 1993). Constraints $\text{ALIGN}(\phi, T^*, \text{Rt})$ (11) and $\text{ALIGN}(\phi, T^*, \text{Left})$ (12) effectively require accents at both edges of a ϕ .

(11) $\text{ALIGN}(\phi, T^*, \text{Rt})$: The right edge of every ϕ coincides with a pitch accent

(12) $\text{ALIGN}(\phi, T^*, \text{Left})$: The left edge of every ϕ coincides with a pitch accent

NoCLASH will weed out intermediate accents. To demonstrate that the ranking of NoCLASH below $\text{ALIGN}(\phi, T^*, \text{Left})$ and $\text{ALIGN}(\phi, T^*, \text{Rt})$ produces the right results, consider a disyllabic expression like *THREE BOOKS*, which will retain the clashing accents so as to satisfy both alignment constraints, as shown in tableau 14. As will be clear, demoting NoCLASH will lead to *THREE books* or *three BOOKS*, depending on whether right or left alignment ranks higher. In the same way, words like CHINESE, PONTOON, SARDINE survive with two accents in citation pronunciation, one on the foot with main stress and one on the preceding foot. By contrast, in *VERY old MEN*, NoCLASH can be satisfied without disturbing either the left or right alignment of accent.

(13) NoCLASH: Pitch accents are not adjacent

(14)

	ALIGN(ϕ , T*, Rt)	ALIGN(ϕ , T*, Left)	NoCLASH
THREE BOOKS			
 a. THREE BOOKS			*
b. THREE books	*!		
c. three BOOKS		*!	
VERY OLD MEN			
 a. VERY old MEN			
b. VERY old men	*!		
c. very old MEN		*!	

Faithfulness constraints will need to preserve lexical accent distributions. In particular, MAX-IO(Accent) (10) keeps the deletion of accents at bay, just as DEP-IO(Accent) (9) will prevent the addition accents. The characterization of the correct candidate for *Tom Paine Big Band* is shown in tableau (15): only the version with accents on the first and last words satisfies the alignment constraints and NoCLASH. Clearly, MAX-IO(Acc) must rank below NoCLASH for the latter constraint to do its work.

(15)

	ALIGN(ϕ , T*, Rt)	ALIGN(ϕ , T*, Left)	NoCLASH	MAX-IO(Acc)
TOM PAINE BIG BAND				
 a. TOM paine big BAND				**
b. TOM paine BIG BAND			*!	*
c. tom PAINE big BAND		*!		**
d. tom paine big BAND		*!*		***
e. TOM paine BIG band	*!			**

Variation in rhythmic accent deletions can be modelled by assuming that MAX-IO(Accent) is variably ranked with constraints that penalize the occurrence of accents that are only one unaccented syllable apart. When *Tom* is replaced with disyllabic *Tomas*, the deletion of the medial accent on *Paine* would depend on the ranking of MAX-IO(Accent) and a constraint that is more strict than NOCLASH: NOREMOTECLASH, which bans [** . **] and is universally ranked below NOCLASH.

- (16) NOREMOTECLASH: Accents are more than one unaccented syllable apart.

If MAX-IO(Accent) $\gg$ NOREMOTECLASH, the pronunciation would be *TOMas PAINE Big BAND*, while the opposite ranking would give *TOMas Paine Big BAND*. Indeed, we might assume a further constraint NOVERYREMOTECLASH, penalizing [** . . **]. If it were ranked above MAX-IO(Accent), medial deaccentuation would occur in *JERemy Paine's Big BAND*. A further application of this interpretation of clash resolution is given in Gussenhoven (2004, ch. 13).

Next, to obtain *TOMcat-free ROOF*, we need to expose the input *TOMcat-FREE ROOF* to the strictures of NOCLASH for the accent of *free* to disappear. Here, the role of DEP-IO(Accent) (9) is made explicit by the inclusion of candidate (d). To illustrate its role further, tableau (17) also shows the case of *Tom PAINE Street BLUES*, where DEP-IO(Accent) prevents the re-accentuation of *Tom*, as in candidate (b).

(17)

	Dep-IO(Accent)	ALIGN(ϕ , T*, Rt)	ALIGN(ϕ , T*, Left)	NOCLASH	MAX-IO(Accent)
TOM cat FREE ROOF					
☞ a. TOM cat free ROOF					*
b. TOM cat FREE ROOF				*!	
c. tom cat FREE ROOF			*!*	*	*
d. tom CAT free ROOF	*!		*		**
tom PAINE street BLUES					
☞ a. tom PAINE street BLUES			*		
b. TOM paine street BLUES	*!				*

Now that the basic mechanism for rhythmic distributions of pitch accents has been illustrated, I turn to data involving bracketing effects.

2.2. Bracketing effects

As argued in Prince (1983), Selkirk (1984) and Gussenhoven (1987) on the basis of examples like (2), rhythmic adjustments in complex NPs unmistakably reveal the effects of the constituent structure. However, the Lexical Phonology version of OT does not allow for post-lexical cyclicity: it rather assumes a single post-lexical module. In section 2.3, I will argue that this assumption is correct, and that a cyclic treatment is in fact inadequate. Instead, the prosodic structure for NP's must allow procliticized ϕ s. The difference between these approaches for a structure like *ethnic Chinese support* is illustrated in (18). A cyclic treatment would deal with the subconstituents *ethnic Chinese* and *support* separately (cycle 1), pass the output on to the cycle 2, where the complete structure is dealt with, as shown in (18a). By contrast, a treatment based on the procliticized structure (18b) deals with the entire structure at once. Note, however, that this structure lacks the internal right-hand ϕ boundary which *is* present in (1a).

- (18) a. ϕ [ethnic-Chinese] ϕ , ϕ [support] ϕ cycle 1
 ϕ [ethnic Chinese support] ϕ cycle 2
 Cyclic treatment
 b. ϕ [ethnic Chinese ϕ [support] ϕ] ϕ
 Procliticized structure

In many structures, the two approaches produce the same, correct output. Tableau (19) deals with *with ETHnic ChiNESE supPORT* cyclically. The first cycle correctly gets rid of the medial accent in *Chi-* through the services of NoCLASH. The adjusted form then combines with *supPORT* to form a phrase on the second cycle, where no further adjustments are needed. Removal of the medial accent on *-nese* is uncalled for, as shown by candidate (e), while incorrect accentuation of *Chi-*, as in candidate (f), is prevented by DEP-IO(Accent), this syllable being unaccented in the input to the second cycle.

(19)

	DEP-IO(Accent)	ALIGN(ϕ , T*, Rt)	ALIGN(ϕ , T*, Left)	NoCLASH	MAX-IO(Accent)	NoREMOTECLASH
[ETHNIC CHINESE]						
☞ a. ETHnic chinese					*	
b. ETHnic CHINESE				!		*
c. ETHnic CHINESE		!			*	*
[with ETHNIC chinese support]						
☞ d. with ETHNIC chinese SUPPORT						*
e. with ETHNIC chinese support					!	
f. with ETHNIC CHINESE support	!				*	

Turning to *with GROWing CHInese supPORT*, we see in tableau (20) that on the first cycle NoCLASH correctly removes the medial accent on *-nese* in *CHInese supPORT*, as in candidate (a). Candidate (b) fails to satisfy NoCLASH and candidate (c) is ruled out by ALIGN(ϕ ,T*,Left). On the second cycle, there is no problem deriving the end product, candidate (d). In particular, we cannot reaccent *-nese*, as in candidate (f), due to DEP-IO(Accent), and cannot gratuitously delete the accent on *Chi-*, as in candidate (e). Candidate (e), a possible form, can be derived by reranking MAX-IO(Accent) and NoREMOTECLASH. This ranking would cause the medial accent on *Chi-* in (2b) to disappear, annihilating the intonational difference between (2a) and (2b), as may indeed happen in less formal speech.

(20)

	DEP-IO(Accent)	ALIGN(ϕ ,T*,Rt)	ALIGN(ϕ ,T*,Left)	NoCLASH	MAX-IO(Accent)	NoREMOTECLASH
[CHINESE support]						
a. CHINESE support					*	
b. CHINESE support				!*		
c. CHINESE support			!*		*	*
[with GROWing CHINESE support]						
d. with GROWing CHINESE support						*
e. with GROWing CHINESE support					!*	
f. with GROWing CHINESE support	*!				*	*

We may even derive the complex (21), originally due to Janet Pierrehumbert (Prince 1983; Selkirk 1984: 194). First observe that if the expression is a single ϕ , the correct output cannot be obtained, since a number of alternative pronunciations are predicted, at least one of which is ungrammatical. Specifically, while (21a) and (21b) are possible, (21c) is not. However, in terms of the severity of the clashes, (21a) equals impossible (21c). A cyclic treatment can obtain the correct result by dealing with (21d) first so as to delete the accent on *Brooke* through NoCLASH (and with (21e), but there is no action here, since *station* already lost its accent in the lexicon through CR), and by considering the whole phrase on the second cycle. On this higher cycle, the accent on *Park* disappears if NoREMOTECLASH $\gg$ MAX-IO (Accent), to give (21b). However, impossible candidate (21c) is unobtainable.

- (21) a. ALEwife Brooke PARKway SUBway station
 b. ALEwife Brooke Parkway SUBway station
 c. ??ALEwife BROOKE Parkway SUBway station
 d. ALEwife Brooke PARKway (from ALEwife BROOKE PARKway)
 e. SUBway station

One of the ways in which OT has been equipped to deal with cyclicity is by means of Output-Output faithfulness constraints in a monocyclic grammar (Kager 1999: ch. 6; McCarthy 2002). Output-output faithfulness relates the pronunciation of a form to the pronunciation of a paradigmatically related form. In our case, we would need a constraint FAITH-OO(ACCENT) (22), which requires that, say, *Alewife Brooke Parkway* in *Alewife Brooke Parkway Subway Station* is pronounced as (21d): *ALEwife Brooke PARKway*. A cyclic effect is created, because of the selection of the candidate that most closely corresponds to the citation pronunciation of the internal constituent. Being identical to it, candidate (a) is closer to (21d) than is candidate (b), which violates FAITH-OO(Accent) twice for having lost an accent on *Parkway* and gained one on *Brooke*. While also violating FAITH-OO(Accent) more than does candidate (a), candidate (c) is ruled out by the extra violation of higher-ranking MAX-IO(Accent). As will be clear, all candidates fare equally badly on (ALIGN)(ϕ, T^*, Rt) for failing to have accents on the feet *way* and *station*.

- (22) FAITH-OO(Accent): The accentuation of a subconstituent in XP' is identical to that of a citation pronunciation of that subconstituent

(23)

	DEP-IO(Accent)	ALIGN(ϕ, T^*, R)	ALIGN($\phi, T^*, Left$)	NoCLASH	MAX-IO(Accent)	NoREMOTECLASH	FAITH-OO(Accent)
O: ALEwife brooke PARKway ALEwife BROOKE PARKway SUBway station							
a. ALEwife brooke PARKway SUBway station		**			*	*	
b. ALEwife BROOKE parkway SUBway station		**			*	*	*!*
c. ALEwife brooke parkway SUBway station		**			***		*

2.3. Why post-lexical cyclicity does not work

Correct as the results obtained in the previous section may seem, this postlexical cyclic treatment cannot be maintained. This can be shown by creating a clash that straddles the boundary between the premodifier and the head. For instance, if we were to replace *PARKway* in (21a)

with *PARK*, a clash would arise between *PARK* and *SUB-*. The problem is that in *ALEwife Brooke PARK SUBway station* only the accent on *PARK* should be deleted, not that on *SUB-*. The problem does not specifically target final accents in the ϕ : we cannot use $\text{ALIGN}(\phi, T^*, \text{Rt})$ to generate the correct form. To show that the problem also applies to non-final accents, a ϕ -final *East* has been added to the example in tableau (24) to provide a rightmost accent which is not involved in the action.

(24)

	IDENT-OO(Accent)	NOREMOTECLASH	MAX-IO(Accent)	NOCLASH	ALIGN(ϕ, T^*, Left)	ALIGN(ϕ, T^*, Rt)	DEP-IO(Accent)
O ₁ : ALEwife brooke PARK O ₂ : SUBway station EAST ALEwife BROOKE PARK SUBway station EAST							
¹⁰⁰ a. ALEwife brooke PARK subway station EAST			**				
¹⁰⁰ b. ALEwife brooke park subway station EAST			**				
c. ALEwife brooke PARK SUBway station EAST			*	!			

Both candidates (a) and (b) fare equally badly on MAX-IO(Accent). Constraint IDENT-OO(Accent) must evaluate both Output₁ and Output₂, and will find that candidate (a) violates it on the basis of the first output and that candidate (b) violates it on the basis of the second. The problem, then, is that a cyclic treatment cannot tell which of the clashing accents *PARK* and *SUB-* is to be deleted, and that therefore the correct candidate (b) cannot be obtained. But the cyclic solution does not merely suffer from indecision, as in (24); it may select the wrong candidate, as in the case of (25). Pronunciation (25a) represents a formal speech style, one that preserves accents that are flanked by at least one unaccented syllable on each side. However, a cyclic treatment would produce (25b), as shown in the cyclic tableau (26). On the first cycle, candidate (a) is selected, since both (c) and (d) violate $\text{ALIGN} T^*, \phi, \text{Left}$, while candidate (b) unnecessarily violates MAX-IO(Acc). On the second cycle, a clash arises on *Ten Jap-*, causing candidate (e), i.e. (25b), to be incorrectly selected as the winner.³

- (25)
- a. TEN JAPaNESE conSTRUCTIONS
b. ??TEN JapaNESE conSTRUCTIONS

³ It is important to see that (25b) is a non-neutral pronunciation of the phrase. This form is appropriate if all three words are pronounced as separate ϕ s, a pronunciation that would be used contrastively with *Eleven Australian demolitions*, when all elements have corrective focus, but this is not the intended reading.

(26)

	Dep-IO(Accent)	ALIGN(ϕ , T*, R)	ALIGN(ϕ , T*, Left)	NoCLASH	MAX-IO(Accent)	NoREMOTECLASH
[JAPANESE CONSTRUCTIONS]						
☞ a. JAPANESE CONSTRUCTIONS						**
b. JAPANESE CONSTRUCTIONS					*!	
c. JAPANESE CONSTRUCTIONS			*!		*	*
d. JAPANESE CONSTRUCTIONS			*!		**	
TEN JAPANESE CONSTRUCTIONS						
☞ ! e. TEN JAPANESE CONSTRUCTIONS					*	*
f. TEN JAPANESE CONSTRUCTIONS				*!		**

Neither is (27a) available under a cyclic treatment. If we rerank NoREMOTECLASH and MAX-IO(Accent), as shown in tableau (28), *JAPANESE conSTRUCTIONS* will be the input to the second cycle, and NoCLASH will select candidate (a), i.e. (27b): on this second cycle, the accent on *Jap-* will again disappear.

- (27) a. TEN JAPANESE conSTRUCTIONS
 b. ??TEN JAPANESE conSTRUCTIONS

(28)

	Dep-IO(Accent)	ALIGN(ϕ , T*, R)	ALIGN(ϕ , T*, Left)	NoCLASH	NoREMOTECLASH	MAX-IO(Accent)
TEN JAPANESE CONSTRUCTIONS						
☞ a. TEN JAPANESE CONSTRUCTIONS						*
b. TEN JAPANESE CONSTRUCTIONS				*!		

Likewise, the result for *Fifteen Japanese constructions* would be the unlikely (29b) under this ranking, again, to the exclusion of the entirely natural (29a). On the first cycle, *-nese* loses its accent, and on the second *Jap-*. If we reverse the ranking of NoREMOTECLASH and MAX-IO(Acc), both syllables will remain accented, as in (29c), since neither violates NoCLASH. However, the point is that if an accent is maintained on *Japanese* at all, it will be the one on *Jap-*: the cyclic treatment is incapable of explaining this.

- (29) a. FIFteen JAPANEse conSTRUCTIONS
 b. ?FIFteen Japanese conSTRUCTIONS
 c. FIFteen JAPANESE conSTRUCTIONS

Example (29a) also demonstrates that we cannot adopt a solution that assigns a ϕ to the embedding pre-modifier. The structure $\phi[FIFTEEN]_{\phi} \phi[JAPANESE conSTRUCTIONS]_{\phi}$ will not allow the final accent in the ϕ *FIFTEEN* to be deleted, contrary to fact, due to high-ranking ALIGN (ϕ, T^*, Rt). While a ϕ -boundary after *Fifteen* would correctly preserve the accent on *Jap-*, it would thus incorrectly preserve the accent on *-teen*. The failure is not restricted to numerals, but is quite general, as illustrated by the pronunciation of *HIGH-TECH* in *HIGH-tech JAPANEse conSTRUCTIONS*, where the accent on *tech*, but not that on *Jap-*, is typically deleted.

The next section argues that the bias towards the retention of the constituent-initial accents and deletion of the constituent-final accents is to be found in a procliticized ϕ -structure.

2.4. Procliticized ϕ s

The asymmetry of NP-internal clash resolutions would be reproduced in a structure that can have a left-hand ϕ -edge without at the same time creating a righthand ϕ -edge, so that the working of ALIGN(ϕ, T^*, Rt) can be suspended internally in the ϕ . This can be done by adopting a proclitic prosodic structure. Specifically, I assume that ϕ 's can be multiply nested as in (30a) and (30b). This represents a minimal deviation from Selkirk's (1995) Non-recursivity. It will have the effect of preserving accents in the locations of the arrows thanks to ALIGN($\phi, T^*, Left$). However, by not having right-hand ϕ -boundaries before these syllables, we make ALIGN(ϕ, T^*, Rt) ineffective, and so allow the accents in the immediately preceding syllables of (30) to disappear. (The accent on *Ten* in (30a) will be preserved by ALIGN($\phi, T^*, Left$), of course.)

- (30) a. $\phi[Ten \ \phi[Japanese \ constructions]_{\phi}]_{\phi}$
 ↑ ↑
 b. $\phi[Twenty-six \ \phi[very \ nice \ \phi[Japanese \ constructions]_{\phi}]_{\phi}]_{\phi}$
 ↑ ↑ ↑

The mechanics of creating these procliticized structures involve a number of elements. First, there is the question of just what morpho-syntactic constituent we are dealing with in the case of internal constituents in multiply pre-modified structures like (30). Second, we explicitly need both right and left alignment of ϕ 's, and third, a way of favouring left-alignment. As for the constituency of the internal constituent, this is taken to be a super-Noun, N' (also: Adjective Phrase: AP), because they do not allow determiners, unlike the

superordinate constituent, the NP (also: a Determiner Phrase, DP), (Radford 1981: 95, Abney 1987). Such internal constituents, like *Japanese constructions* in (30), can be provided with ϕ 's if we assume that XP in $\text{ALIGN}(\text{XP}, \phi)$ refers to the internal constituents (regardless of whether we are dealing with an Adjective Phrase, a Numeral, or a high N'), so that XP' in $\text{ALIGN}(\text{XP}', \phi)$ can be reserved to make reference to the maximal constituent. Second, to be able to manipulate edges separately, we need both $\text{ALIGN}(\text{XP}, \phi, \text{Rt})$, given in (31), and $\text{ALIGN}(\text{XP}, \phi, \text{Left})$, given in (32). And third, to create the asymmetric bracketing, $\text{ALIGN}(\text{XP}', \phi, \text{Rt})$ will need to rank above, but $\text{ALIGN}(\text{XP}, \phi, \text{Rt})$ below a constraint forbidding right-hand ϕ -boundaries, given as $*\phi, \text{Rt}$ in (34). This latter constraint is one half of Truckenbrodt's (2002) $*P$, which forbids ϕ 's altogether. Its counterpart obviously is $*\phi, \text{Left}$ (35).

- (31) $\text{ALIGN}(\text{XP}', \phi, \text{Rt})$: Align the right edge of XP' with the right edge of ϕ .
- (32) $\text{ALIGN}(\text{XP}, \phi, \text{Left})$: Align the left edge of every XP with the left edge of ϕ .
- (33) $\text{ALIGN}(\text{XP}, \phi, \text{Rt})$: Align the right edge of every XP with the right edge of ϕ .
- (34) $*\phi, \text{Rt}$: Do not have a right edge of ϕ .
- (35) $*\phi, \text{Left}$: Do not have a left edge of ϕ .

Tableau (36) shows the intended syntax-phonology mapping. Notice that constraints (34) and (35) can only be violated once at any edge, as shown for $*\phi, \text{Rt}$ in candidate (a), where there are two brackets after *constructions* which constitute a single edge. Candidate (b) fails to have a final right-hand boundary at all, candidate (c) has one too many, while candidate (d) lacks the required internal left-hand boundary.

(36)

	$\text{ALIGN}(\text{XP}, \phi, \text{Rt})$	$*\phi, \text{Rt}$	$\text{ALIGN}(\text{XP}, \phi, \text{Rt})$	$\text{ALIGN}(\text{XP}, \phi, \text{Left})$	$*\phi, \text{Left}$
FIFTEEN JAPANESE CONSTRUCTIONS					
a. $\emptyset[\text{FIFTEEN } \emptyset[\text{JAPANESE CONSTRUCTIONS}] \emptyset] \emptyset$		*	*		**
b. $\emptyset[\text{FIFTEEN JAPANESE CONSTRUCTIONS...}]$	*!		**	*	*
c. $\emptyset[\text{FIFTEEN}] \emptyset \emptyset[\text{JAPANESE CONSTRUCTIONS}] \emptyset$		**!			**
d. $\emptyset[\text{FIFTEEN JAPANESE CONSTRUCTIONS}] \emptyset$		*	*	*!	*

Tableau (37) shows how the proclitic ϕ -structure (36a), with the accent distribution of (29a), wins. Unlike candidates (c) and (d), it fully satisfies $\text{ALIGN}(\phi, T^*, \text{Left})$ by having an accent on *Jap-*. And it is preferred over

candidate (b) by having no accent on *-nese*, thereby avoiding a violation of NoREMOTECLASH which *is* incurred by candidate (b). Reranking of MAX-IO(Accent) and NoREMOTECLASH would identify candidate (b) as the winner, (= (29c)).

(37)

	DEP-IO(Accent)	ALIGN(ϕ , T*, R)	ALIGN(ϕ , T*, L)	NoCLASH	NoREMOTECLASH	MAX-IO(Accent)
[FIFTEEN [JAPANESE CONSTRUCTIONS] $\emptyset$ $\emptyset$						
a. [FIFTEEN [JAPANESE CONSTRUCTIONS] $\emptyset$ $\emptyset$					*	**
b. [FIFTEEN [JAPANESE CONSTRUCTIONS] $\emptyset$ $\emptyset$					**!	*
c. [FIFTEEN [japanese constructions] $\emptyset$ $\emptyset$			*!			***
d. [FIFTEEN [japanese constructions] $\emptyset$ $\emptyset$			*!		*	**

This concludes the analysis. The next section introduces a contrast which has not, as far as I am aware, received any attention. It appears to be correctly dealt with in the new analysis.

2.5. *An apparent counterexample*

The new analysis would appear to create a problem, in that it will not be able to obtain candidate (c) in tableau (37), even though it may appear to be a possible fast-speech pronunciation. Or again, while pronunciations like *TEN JAPANESE conSTRUCTIONS* can now be accounted for, it is not clear how we could ever lose the accent on *Jap-* to create the less formal pronunciation *TEN japanese conSTRUCTIONS*: it'll be stubbornly preserved by ALIGN(XP, ϕ , Left). Worse, the pronunciation predicted for *tired old men* is *TIRED OLD MEN* under the new analysis, not *TIRED old MEN*, since both *tired* and *old* will have a ϕ -boundary on their left.

Before rushing to repair this apparent flaw in the analysis, a closer look at the data is required. A subtle prosodic difference between accented and unaccented syllables that are flanked by accented syllables supports the analysis. In particular, observe that while (38a,b) are prosodically identical, (39a,b) are not.

- (38)
- a. FOURTEEN
- b. FOUR TEENS (i.e. teenagers)

- (39)
- a. FOURteen BOYS
- b. FOUR KEEN BOYS

If *four keen boys* is pronounced with three accents, as in (39b), the middle accent will be least prominent, but this pattern is distinct from one in which the medial syllable is unaccented, as in (39a). The two-accent

pattern occurs in left-branching $[[A\ B]\ C]$, like *THREE ton ROCK, LONG haul TRUCK, NORTH Sea OIL*; it would also be expected in right-branching $[A\ [B\ C]]$ when B is unaccented for independent, pragmatic reasons, such as when *old MEN* is premodified with *TIRED*. The usual pronunciation of the right-branching type, however, is with three accents, as in *SIX GREEK MEN, ONE BIG MESS, ALL FIVE BOOKS*, etc. In this connection, Eric Kellerman provided the minimal pair in (40), which is particularly significant because the less common expression (40a) obligatorily loses the medial accent, while (40b) does not, and thus illustrates the structural, non-pragmatic basis of these patterns.

- (40) a. TOO tall BUILDings
b. TWO TALL BUILDings

Having defended the analysis against the possible criticism that NoCLASH must be active in $[A\ [B\ C]]$ structures, we now need to answer the question how, in more rapid speech, the NP-internal ϕ -initial accent could ever be deleted at all. Arguably, procliticized structures may be restructured to single ϕ s in less formal speech. With reference to the bracketing of *FIFTEEN JAPANESE conSTRUCTIONS* in tableau (36), it is easy to see that a reranking of $\text{ALIGN}(XP, \phi, \text{Left})$ and $*\phi, \text{Left}$ will select candidate (d), the single ϕ , over candidate (a), the procliticized ϕ , making candidate (d) the winner.

3. Conclusion

The distribution of pitch accents in English is determined by the interaction of lexical accent rules, such as the Compound Rule and Initial Accent Deletion, and postlexical rhythmic readjustments. Just as in French (Post 2000) and Italian (Nespor & Vogel 1986: 175), the latter are due to clash avoidance within the ϕ . In order to account for the English data, in particular those involving multiple premodifying adjectives, a ϕ -structure with procliticization needed to be assumed. I considered and rejected analyses which employ (a) cyclicity, (b) single, restructured ϕ , or (c) separate ϕ s. Proclitic structures are the only way in which we could systematically delete the final accent in pre-modifying adjectives and preserve the initial accent in the pre-modified NP, as in *TWENTY-six CHINESE conSTRUCTIONS* (**TWENTY-SIX CHINESE conSTRUCTIONS*).

The solution incorporates the earlier explanation for the difference between (41a,b), originally due to (Hayes 1984), which Gussenhoven (1991) attributed to a prosodic restructuring of a left-branching to a right-branching structure. Thus, (41a) contains a proclitic *hundred*, allowing both *hun-* and *thir-* to retain their accents. Example (41b) consists of two ϕ s, since the clash between *-teen* and *Main* remains

unrelieved. (All stress marks in these examples indicate accented syllables.)

- (41) a. ϕ [hùndred ϕ [thírtēen mén] ϕ] ϕ (i.e., 113 men)
 b. ϕ [hùndred thirtēen] ϕ ϕ [Máin Street] ϕ

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